

This theorem is usually evoked to justify the density of polynomials when $X \subset \mathbb{R}^d$. Trigonometric polynomials are also dense for the same reason (and, by the way, this may be used to prove the L^2 convergence of the Fourier series). In product spaces $X = Y \times Z$, linear combinations of separable functions of the form $\sum_i \varphi_i(y) \psi_i(z)$ are dense because of this very same theorem.

We just proved that the map T is a transport map between μ and ν . Moreover, by construction, should it be a regular map, the Knothe transport T has a triangular Jacobian matrix with nonnegative entries on its diagonal. Compare this with the Brenier optimal map for the quadratic cost: its Jacobian matrix is the Hessian matrix of a convex function, which is symmetric (instead of triangular) and has also positive eigenvalues. Moreover, we underline that assuming absolute continuity as we did, all the monotone optimal transports that we use in defining T are invertible, since not only the source measures are atomless, but also the arrival ones are atomless (they are all absolutely continuous). As a consequence, each map $\hat{T}^k$ is also invertible.

Remark 2.19. It is interesting to note that, due to its explicit definition, the Knothe map automatically inherits some regularity properties from the measures μ and ν . In general, T has the same regularity as μ and ν and not better. Indeed, each T^k is built as a monotone transport, which gives one extra derivative than the regularity of μ and ν (see Remark 2.7), but only in the direction of the variable x_k . With respect to the other variables, the explicit formula of monotone transports with cumulative distribution functions allows to prove that the dependence on $(x_{k+1}, \dots, x_d)$ is as regular as the densities are.

Remark 2.20. It is also interesting to note that the Knothe map defined above is the (unique) transport map from μ to ν which is monotone for the lexicographic order, i.e. the order where $x < y$ is defined in the following way: “there exists $k \in \{1, 2, \dots, d\}$ such that $x_j = y_j$ for $j > k$ and $x_k < y_k$ ”.

Remark 2.21. How to define the Knothe transport for μ and ν which are not absolutely continuous? The construction uses the notion of disintegration of measures (Box 2.2).

First we define the transport map $T^d : \mathbb{R} \rightarrow \mathbb{R}$ as the monotone non-decreasing transport map sending $(\pi_d)_\# \mu$ onto $(\pi_d)_\# \nu$. We need to suppose that $(\pi_d)_\# \mu$ has no atoms. Then, for each k we consider $(\pi_{k,d})_\# \mu$ and we disintegrate it with respect to $(\pi_{k+1,d})_\# \mu$, and we do the same for $(\pi_{k,d})_\# \nu$ with $(\pi_{k+1,d})_\# \nu$. We obtain some families of probabilities on the real line that we can call $\mu_{x_{k+1}, \dots, x_n}^k$ and $\nu_{y_{k+1}, \dots, y_n}^k$, respectively. They are measures on the variables x_k and y_k and they replace the densities $f^k(\cdot, x_{k+1}, \dots, x_d)$ and $g^k(\cdot, y_{k+1}, \dots, y_d)$. Again, we need to suppose that each measure $\mu_{x_{k+1}, \dots, x_n}^k$ is atomless. The maps $\hat{T}^k(x_k, x_{k+1}, \dots, x_d)$ are defined as before, using one-dimensional monotone increasing maps.

2.4 Knothe transport as a limit of quadratic optimal transports

Let us slightly modify the quadratic cost that we discussed in Section 1.3.1 and replace it with the weighted quadratic cost

$$c_\varepsilon(x, y) := \sum_{i=1}^d \lambda_i(\varepsilon) (x_i - y_i)^2$$

where the $\lambda_i(\varepsilon)$'s are positive scalars depending on a parameter $\varepsilon > 0$. If μ is absolutely continuous with respect to the Lebesgue measure, the corresponding optimal transportation problem admits a unique solution T_ε . For simplicity, in this section we will only consider the case where the supports of μ and ν are compact (but everything could be adapted to the case $\int |x|^2 d\mu, \int |y|^2 d\nu < +\infty$).

When in addition, for all $k \in \{1, \dots, d-1\}$, $\lambda_k(\varepsilon)/\lambda_{k+1}(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$, it is natural to expect the convergence of T_ε to the Knothe transport T . This convergence result was conjectured by Y. Brenier as a very natural one.

We will show it under the absolute continuity assumption of the previous section. On the other hand, [114] proves the same result under a slightly weaker assumption, namely the absence of atoms in all the disintegrated measures. The curious point is that absence of atoms is also needed on the target measure ν , and that a counter-example to the convergence result exists if it is not satisfied.

Example 2.22 (explicit computations for Gaussians). To illustrate the problem in a particular case where explicit solutions are available, take $d = 2$, and μ and ν two Gaussian measures where $\mu = N(0, I)$ and $\nu = N\left(0, \begin{pmatrix} a & b \\ b & c \end{pmatrix}\right)$ (with $ac > b^2$, $a > 0$; the notation $N(a, A)$ stands for the Gaussian density with covariance matrix A , centered at a). Take $\lambda_1(\varepsilon) = \varepsilon$ and $\lambda_2(\varepsilon) = 1$. Then it can be verified (see also Ex(11)) that T_ε is linear, and that its matrix in the canonical basis of $\mathbb{R}^2$ is

$$T_\varepsilon = \frac{1}{\sqrt{a\varepsilon^2 + c + 2\varepsilon\sqrt{ac - b^2}}} \begin{pmatrix} a\varepsilon + \sqrt{ac - b^2} & b \\ b\varepsilon & c + \varepsilon\sqrt{ac - b^2} \end{pmatrix}$$

which converges as $\varepsilon \rightarrow 0$ to $T = \begin{pmatrix} \sqrt{a - b^2/c} & b/\sqrt{c} \\ 0 & \sqrt{c} \end{pmatrix}$, i.e. the matrix of the Knothe transport from μ to ν .

We directly state our first result, whose proof, in the spirit of Γ -convergence developments (see [134] and Box 4.6), will require several steps.

Theorem 2.23. *Let μ and ν be two absolutely continuous probability measures on $\mathbb{R}^d$ with compact supports and $\gamma_\varepsilon \in \Pi(\mu, \nu)$ be an optimal transport plan for the costs $c_\varepsilon(x, y) = \sum_{i=1}^d \lambda_i(\varepsilon) (x_i - y_i)^2$, for some weights $\lambda_k(\varepsilon) > 0$. Suppose that for all $k \in \{1, \dots, d-1\}$, $\lambda_k(\varepsilon)/\lambda_{k+1}(\varepsilon) \rightarrow 0$ as $\varepsilon \rightarrow 0$. Let T be the Knothe-Rosenblatt map between μ and ν and $\gamma_K \in \mathcal{P}(\mathbb{R}^d \times \mathbb{R}^d)$ the associated transport plan (i.e. $\gamma_K := (\text{id}, T)\# \mu$). Then $\gamma_\varepsilon \rightarrow \gamma_K$ as $\varepsilon \rightarrow 0$.*

Moreover, the plans γ_ε are induced by transport maps T_ε , and these maps converge to T in $L^2(\mu)$ as $\varepsilon \rightarrow 0$.

The proof will roughly stick to the following strategy. We will take a limit point of γ_ε and show that it shares the same (x_d, y_d) -marginal as γ_K . Then, we will disintegrate with respect to (x_d, y_d) and prove that the conditional (x_{d-1}, y_{d-1}) -marginals coincide. We will end the proof by iterating this argument.

Proof. Without loss of generality, we assume $\lambda_d(\varepsilon) = 1$ and $\lambda_i(\varepsilon)/\lambda_{i+1}(\varepsilon) \rightarrow 0$.

Take the optimal plans γ_ε and suppose (which is possible, up to subsequences) $\gamma_\varepsilon \rightharpoonup \gamma$. We want to prove $\gamma = \gamma_K$. From the uniqueness of the Knothe transport, the convergence will hold on the full sequence, and not only on subsequences.

By comparing γ_ε to γ_K and using optimality we first get

$$\int c_\varepsilon d\gamma_\varepsilon \leq \int c_\varepsilon d\gamma_K \quad (2.5)$$

and, passing to the limit as $\varepsilon \rightarrow 0$, since c_ε converges uniformly to $c^{(d)}(x, y) = (x_d - y_d)^2$ (we will use the general notation $c^{(k)}(x, y) = |x_k - y_k|^2$), we get

$$\int c^{(d)} d\gamma \leq \int c^{(d)} d\gamma_K.$$

Yet, the function $c^{(d)}$ only depends on the variables x_d and y_d and this shows that the measure $(\pi_d^{x,y})_\# \gamma$ gets a result at least as good as $(\pi_d^{x,y})_\# \gamma_K$ with respect to the quadratic cost ($\pi_d^{x,y}$ being the projection onto the last coordinates of x and y , i.e. $\pi_d^{x,y}(x, y) = (x_d, y_d)$). Yet, the measure γ_K has been chosen on purpose to be optimal from μ^d to ν^d for this cost, and this optimal transport plan is unique. Then $(\pi_d^{x,y})_\# \gamma = (\pi_d^{x,y})_\# \gamma_K$. Let us call γ^d this common measure.

We go back to (2.5) and go on by noting that all the measures γ_ε have the same marginals as γ_K and hence their (separate) projection onto x_d and y_d are μ^d and ν^d , respectively. This implies that $(\pi_d^{x,y})_\# \gamma_\varepsilon$ must realize a result which is not better than $(\pi_d^{x,y})_\# \gamma_K$ as far as the 1D quadratic cost is concerned. Consequently, we have

$$\begin{aligned} & \int |x_d - y_d|^2 d(\pi_d^{x,y})_\# \gamma_K(x_d, y_d) + \sum_{i=1}^{d-1} \lambda_i(\varepsilon) \int (x_i - y_i)^2 d\gamma_\varepsilon \\ & \leq \int c_\varepsilon d\gamma_\varepsilon \leq \int c_\varepsilon d\gamma_K \\ & = \int |x_d - y_d|^2 d(\pi_d^{x,y})_\# \gamma_K(x_d, y_d) + \sum_{i=1}^{d-1} \lambda_i(\varepsilon) \int (x_i - y_i)^2 d\gamma_K, \end{aligned}$$

which implies, by simplifying the common term, dividing by $\lambda_{d-1}(\varepsilon)$ and passing to the limit,

$$\int c^{(d-1)} d\gamma \leq \int c^{(d-1)} d\gamma_K. \quad (2.6)$$

We can note that both integrals depend on the variables x_{d-1} and y_{d-1} only. Anyway, we can project onto the variables (x_{d-1}, x_d) and (y_{d-1}, y_d) (obtaining measures $(\pi_{d-1,d}^{x,y})_{\#}\gamma$ and $(\pi_{d-1,d}^{x,y})_{\#}\gamma_K$) so that we disintegrate with respect to the measure γ^d . We have

$$\begin{aligned} \int d\gamma^d(x_d, y_d) \int |x_{d-1} - y_{d-1}|^2 d\gamma_{(x_d, y_d)}^{d-1}(x_{d-1}, y_{d-1}) \\ \leq \int d\gamma^d(x_d, y_d) \int |x_{d-1} - y_{d-1}|^2 d\gamma_{(x_d, y_d), K}^{d-1}(x_{d-1}, y_{d-1}). \end{aligned} \quad (2.7)$$

We want to prove that the measures $\gamma_{(x_d, y_d)}^{d-1}$ share the same marginals on x_{d-1} and y_{d-1} of the corresponding $\gamma_{(x_d, y_d), K}^{d-1}$. Should this be the case, their quadratic cost would be not better than the corresponding cost of $\gamma_{(x_d, y_d), K}^{d-1}$ (because the Knothe measure has been chosen exactly with the intention of being quadratically optimal on (x_{d-1}, y_{d-1}) once x_d and y_d are fixed). Yet, (2.7) shows that, on average, the result given by those measures is not worse than the results of the optimal ones. Thus, the two results coincide for almost any pair (x_d, y_d) and, by uniqueness of the optimal transports in the 1D case, we get $\gamma_{(x_d, y_d)}^{d-1} = \gamma_{(x_d, y_d), K}^{d-1}$. To let this proof work it is sufficient to prove that the marginals of the two measures coincide for γ^d -a.e. pair (x_d, y_d) . For fixed (x_d, y_d) we would like to prove, for any ϕ

$$\int \phi(x_{d-1}) d\gamma_{(x_d, y_d)}^{d-1} = \int \phi(x_{d-1}) d\gamma_{(x_d, y_d), K}^{d-1}$$

(and to prove an analogous equality for functions of y_{d-1}). Since we want to prove it for a.e. pair (x_d, y_d) , it is sufficient to prove this equality:

$$\begin{aligned} \int d\gamma^d(x_d, y_d) \psi(x_d, y_d) \int \phi(x_{d-1}) d\gamma_{(x_d, y_d)}^{d-1} \\ = \int d\gamma^d(x_d, y_d) \psi(x_d, y_d) \int \phi(x_{d-1}) d\gamma_{(x_d, y_d), K}^{d-1} \end{aligned}$$

for any ϕ and any ψ . This means proving

$$\int \psi(x_d, y_d) \phi(x_{d-1}) d\gamma^{d-1} = \int \psi(x_d, y_d) \phi(x_{d-1}) d\gamma_K^{d-1},$$

which is not trivial. As far as now, we only know that the two measures γ^{d-1} and γ_K^{d-1} have the same marginals with respect to the pairs (x_{d-1}, x_d) , (y_{d-1}, y_d) (since they have the same projections onto x and onto y) and (x_d, y_d) (since we just proved it). But here there is a function of the three variables (x_{d-1}, x_d, y_d) . Yet, we know that the measure γ^d is concentrated on the set $y_d = T_d(x_d)$ for a certain map T_d (here we use that the optimal plan is indeed a map), and this allows to replace the expression of y_d , thus getting rid of one variable. This proves that the function $\psi(x_d, y_d) \phi(x_{d-1})$ is actually a function of (x_{d-1}, x_d) only, and that equality holds when passing from γ to γ_K . The same can be performed on functions $\psi(x_d, y_d) \phi(y_{d-1})$ but we have in this case to ensure that we can replace x_d with a function of y_d , i.e. that we can invert T_d . This is possible thanks to

the absolute continuity of ν^d , since T_d is the optimal transport from μ^d to ν^d , but an optimal transport exists in the other direction as well and it gives the same optimal plan (thanks to uniqueness). These facts prove that the measures $\gamma_{(x_d, y_d)}^{d-1}$ and $\gamma_{(x_d, y_d), K}^{d-1}$ have the same marginals and hence, since they are both optimal, they coincide for a.e. pair (x_d, y_d) . This implies $(\pi_{d-1, d}^{x, y})\# \gamma = (\pi_{d-1, d}^{x, y})\# \gamma_K$, and we will call this common measure γ^{d-1} .

In the case $d = 2$, the proof would be finished. Now, it is possible to go on by induction in the following way: we prove, by induction on N , that the statement of the theorem is true if $d \leq N$ and that, whatever the dimension is, we have $(\pi_{k, d}^{x, y})\# \gamma = (\pi_{k, d}^{x, y})\# \gamma_K$ (i.e. that we have convergence in what concerns the components from the k -th to the d -th) for every $k \geq d - (N - 1)$. Both facts are proven for $N = 2$.

Suppose the result proven for a certain value of N . Take $d > N$ and $k = d - (N - 1)$. We want to prove $(\pi_{k-1, d}^{x, y})\# \gamma = (\pi_{k-1, d}^{x, y})\# \gamma_K$. If we can do it, we are done, since this also proves that the result is true in dimension $d = N + 1$ (since in this case we have $k = 2$ and with our extra step we conclude the proof).

Write $x = (x^-, x^+)$ and $y = (y^-, y^+)$ with $x^-, y^- \in \mathbb{R}^{k-1}$ and $x^+, y^+ \in \mathbb{R}^{d-k+1}$ (decomposing into the components up to $k-1$ and after k). Then, define a competitor γ_K^ε in the following way: γ_K^ε is a measure on $\mathbb{R}^d \times \mathbb{R}^d$ with the following marginals

$$(\pi_{x^-, x^+})\# \gamma_K^\varepsilon = \mu, (\pi_{y^-, y^+})\# \gamma_K^\varepsilon = \nu, (\pi_{x^+, y^+})\# \gamma_K^\varepsilon = \eta_{\varepsilon, k},$$

where $\eta_{\varepsilon, k}$ is the measure which optimizes the transport cost $\sum_{i=k}^d \lambda_i(\varepsilon) c^{(i)}$ between the marginals $\mu^k = (\pi_{x^+})\# \mu$ and $\nu^k = (\pi_{y^+})\# \nu$. Thanks to the recursive structure of the proof, we know that $\eta_{\varepsilon, k}$ converges to $(\pi_{x^+, y^+})\# \gamma_K$ (since we suppose the result to be true in dimension N).

We need to specify how to build the measure γ_K^ε (which is likely not to be unique). One easy way to construct such a measure is the following: disintegrate μ and ν according to π_{x^+} and π_{y^+} , respectively, thus getting two family of measures μ^{x^+} and ν^{y^+} . Then, for every pair (x^+, y^+) pick a transport plan γ^{x^+, y^+} between them. The measure $\gamma^{x^+, y^+} \otimes \eta_{\varepsilon, k}$ satisfies the required properties. Moreover, we will choose for γ^{x^+, y^+} the Knothe transport between those two measures. Indeed, for the sake of this step of the proof, the only important point is that, in what concerns the $(k-1)$ -th component, we choose for any (x^+, y^+) the monotone map from $\mu_{k-1}^{x^+} := (\pi_{x_{k-1}})\# \mu^{x^+}$ to $\nu_{k-1}^{y^+} := (\pi_{y_{k-1}})\# \nu^{y^+}$, so that $\int c^{(k-1)} d\gamma^{x^+, y^+} = \mathcal{T}_c(\mu_{k-1}^{x^+}, \nu_{k-1}^{y^+})$ (in this last expression $\mathcal{T}_c$ denotes the quadratic transport cost on the real line).

With this choice in mind, we write

$$\begin{aligned} & \sum_{i \geq h} \lambda_i(\varepsilon) \int |x_i - y_i|^2 d\gamma_K^\varepsilon + \sum_{i=1}^{k-1} \lambda_i(\varepsilon) \int (x_i - y_i)^2 d\gamma_\varepsilon \\ & \leq \int c_\varepsilon d\gamma_\varepsilon \leq \int c_\varepsilon d\gamma_K^\varepsilon \\ & = \sum_{i \geq h} \lambda_i(\varepsilon) \int |x_i - y_i|^2 d\gamma_K^\varepsilon + \sum_{i=1}^{k-1} \lambda_i(\varepsilon) \int (x_i - y_i)^2 d\gamma_K^\varepsilon, \end{aligned}$$

and consequently, by erasing the common terms and dividing by $\lambda_{k-1}(\varepsilon)$, we get

$$\int c^{(k-1)} d\gamma_\varepsilon \leq \int c^{(k-1)} d\gamma_K^\varepsilon = \int \mathcal{T}_c(\mu_{k-1}^{x^+}; \nu_{k-1}^{y^+}) d\eta_{\varepsilon,h}.$$

Then we pass to the limit as $\varepsilon \rightarrow 0$. The left-hand side of the last inequality tends to $\int c^{(k-1)} d\gamma$, while the right-hand side converges to

$$\int \mathcal{T}_c(\mu_{k-1}^{x^+}; \nu_{k-1}^{y^+}) d(\pi_{x^+,y^+})_\# \gamma_K = \int c^{(k-1)} d\gamma_K.$$

This convergence result is justified by the following Corollary 2.24 of Lemma 1.8. In the end we get

$$\int c^{(k-1)} d\gamma \leq \int c^{(k-1)} d\gamma_K$$

and we can go on. This last inequality was obtained thanks to a procedure which was similar to what we did for getting (2.6), but it required some more work since γ_K is not exactly optimal for all the components $i \geq k$ as it was the case for $k = d$.

We disintegrate with respect to $\pi_{k,d}^{x,y}$ and we act exactly as before: proving that the marginals of the disintegrations coincide is sufficient to prove equality of the measures. Here we will use test-functions of the form

$$\psi(x_h, x_{k+1}, \dots, x_d, y_h, y_{k+1}, \dots, y_d) \varphi(x_{k-1})$$

and

$$\psi(x_h, x_{k+1}, \dots, x_d, y_h, y_{k+1}, \dots, y_d) \varphi(y_{k-1}).$$

The same trick as before, i.e. replacing the variables y with functions of the variables x is again possible. To invert the trick and replace x with y one needs to invert part of Knothe's transport. This is possible since, as we noted, our assumptions implied that all the monotone transports we get are invertible. In the end we get, as before, $\gamma^{k-1} = \gamma_K^{k-1}$. This proves that we can move from step N to step $(N+1)$ in the induction and allows to conclude.

We have now proven $\gamma_\varepsilon \rightarrow \gamma_K$. Yet, if all these transport plans come from transport maps, then $(\text{id}, T_\varepsilon)_\# \mu \rightarrow (\text{id}, T)_\# \mu$ implies $T_\varepsilon \rightarrow T$ in $L^2(\mu)$ (see Lemma 2.25 below). $\square$ $\square$

In the proof we used the following result, which is a corollary of Lemma 1.8.

Corollary 2.24. *If μ and ν are two probabilities over $\mathbb{R}^d = \mathbb{R}^k \times \mathbb{R}^{d-k}$ with compact support which disintegrate as $\mu = \mu^+ \otimes \mu^{x^+}$ and $\nu = \nu^+ \otimes \nu^{y^+}$, and c is a continuous cost, then the functional*

$$\eta \mapsto \int \mathcal{T}_c(\mu^{x^+}, \nu^{y^+}) d\eta(x^+, y^+)$$

is continuous over the transport plans between μ^+ and ν^+ .

Proof. It is sufficient to apply Lemma 1.8 to the following setting: $X = Y = \mathbb{R}^k$, $\tilde{X} = \tilde{Y} = \mathcal{P}(\Omega)$, where $\Omega \subset \mathbb{R}^{d-k}$ is such that $\text{spt}(\mu), \text{spt}(\nu) \subset \mathbb{R}^k \times \Omega$. The space $\mathcal{P}(\Omega)$ is endowed with any distance metrizing weak convergence and making this space a separable metric space (see, for instance, Chapter 5 for a choice of distances with this property, or build it with standard methods for the weak convergence in the unit ball of a Banach space). For the cost function which appears in the statement of Lemma 1.8 we take $(a, b) \mapsto \mathcal{T}_c(a, b)$ and $a(x^+) := \mu^{x^+}$, $b(y^+) := \nu^{y^+}$. Since this quantity is bounded by $\max |c|$ (as we are on compact supports), there is no difficulty in choosing two bounded functions f, g so that $\mathcal{T}_c(a, b) \leq f(a) + g(b)$. $\square$ $\square$

Another easy lemma that we need is the following.

Lemma 2.25. *Given $\mu \in \mathcal{P}(X)$ and $\nu_n \in \mathcal{P}(\Omega)$, where $\Omega \subset \mathbb{R}^d$ is a compact set, suppose that $\gamma_n := \gamma_{T_n} \in \Pi(\mu, \nu_n)$ is such that $\gamma_n \rightarrow \gamma = \gamma_T$. Then $T_n \rightarrow T$ in $L^2(X, \mu)$.*

Proof. Since the functions T_n are valued in Ω , they are equibounded and they admit a weak limit in $L^2(X, \mu)$ up to subsequences. Let us call S this limit. To prove $S = T$, take a continuous function $\xi \in C(X; \mathbb{R}^d)$ and consider the limit of

$$\begin{aligned} \int (\xi(x) \cdot S(x)) d\mu(x) &= \lim_n \int (\xi(x) \cdot T_n(x)) d\mu(x) = \lim_n \int (\xi(x) \cdot y) d\gamma_n(x, y) \\ &= \int (\xi(x) \cdot y) d\gamma_T(x, y) = \int (\xi(x) \cdot T(x)) d\mu(x). \end{aligned}$$

This proves $S = T$.

In order to prove strong convergence, just consider

$$\int |T_n|^2 d\mu = \int |y|^2 d\gamma_n \rightarrow \int |y|^2 d\gamma = \int |T|^2 d\mu.$$

This proves that we do not only have weak convergence in L^2 , but also convergence of the L^2 norm. This implies strong convergence, as usual in Hilbert spaces. $\square$ $\square$

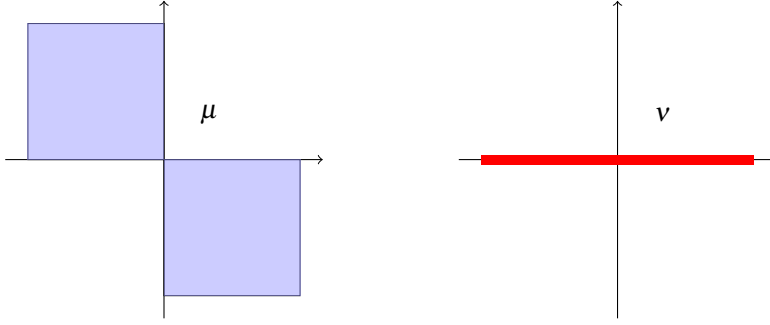
Note that the above proof also works in other spaces L^p and that the boundedness assumption (i.e. compact supports) could be replaced by suitable integrability bounds.

Let us remark here that if instead of considering the quadratic cost c_ε , one considers the more general separable cost

$$c_\varepsilon(x, y) := \sum_{i=1}^d \lambda_i(\varepsilon) c_i(x_i - y_i)$$

where each c_i is a smooth strictly convex function (with suitable growth), then the previous convergence proof carries over.

Example 2.26 (A counterexample when the measures have atoms). We now show that interestingly, the absolute continuity hypothesis in Theorem 2.23 is necessary not only for μ , but also for ν (see Remark 2.21 to check the definition of the Knothe map in such a case). We propose a very simple example in $\mathbb{R}^2$ where μ is absolutely continuous

Figure 2.3: The measures μ and ν in the counter-example

with respect to the Lebesgue measure but ν is not, and we show that the conclusion of the Theorem fails. On the square $\Omega := [-1, 1] \times [-1, 1]$, define $\mu = \frac{1}{2} \mathbb{1}_E \cdot \mathcal{L}^2$, where $E = \{(x_1, x_2) \in [-1, 1] \times [-1, 1] : x_1 x_2 < 0\}$, so that the measure μ is uniformly spread on the upper left and the lower right quadrants, and $\nu = \frac{1}{2} \mathcal{H}_S^1$, S being the segment $[-1, 1] \times \{0\}$.

The Knothe-Rosenblatt map is easily computed as

$$T(x) := (2x_1 + \text{sign}(x_2)), 0)$$

(indeed, the measure ν^2 is a Dirac mass at 0, hence $T^2(x) = 0$). The solution of any transportation problem with cost $|x_1 - y_1|^2 + \varepsilon |x_2 - y_2|^2$ is $T^0(x) := (x_1, 0)$ (no transport may do better than this one, which projects on the support of ν). Therefore, in this example the optimal transportation maps fail to tend to the Knothe-Rosenblatt map. The reason is the atom in the measure $\nu^2 = \delta_0$.

2.5 Discussion

2.5.1 Histogram equalization

A very typical problem where we encounter monotone transport maps in image processing is histogram equalization. The setting is easy to explain: take a black and white digital picture, and consider all possible grey levels; these levels are number from 0 to 255; for each $i = 0, \dots, 255$ count how many pixels have exactly such a grey level, and make an histogram of that. In some situations, it could be interesting to transform the given picture into a different one, which should represent again the same subject, but have a different histogram. Why? a typical example is the following: you have two pictures, A and B, which are supposed to represent the same object, with minor modifications (something has moved. . .), at different instants of time; you want to compare them, but B happens to be much darker than A (this means that the histogram of B is much more concentrated on the right-hand part of the graph than that of A). Then, before comparing them, one prefers to transform B into a new picture B', where the darkness level of B' is comparable to that of A, i.e. the histogram of B' is equal to that